

Supplementary Table 2. Verified confusion and per-class F1 summary

Model	C1 -> CD56bright NK	C1 <-> C2	C2 <-> C3	Naive-like ILC -> C1	CD56dim NK F1	ILC2 F1
PCA + LR	46/600 (7.7%)	38/1200 (3.2%)	41/805 (5.1%)	9/119 (7.6%)	0.980	0.986
scGPT MLP	55/600 (9.2%)	118/1200 (9.8%)	54/805 (6.7%)	27/119 (22.7%)	0.976	0.948
Geneformer MLP	107/600 (17.8%)	193/1200 (16.1%)	97/805 (12.0%)	39/119 (32.8%)	0.872	0.844
UCE MLP	95/600 (15.8%)	85/1200 (7.1%)	53/805 (6.6%)	23/119 (19.3%)	0.974	0.968

Confusion entries are count/true-class support (%). F1 values are from existing classification reports.